MODULE 13 · DOMAIN 2 - CONFIGURATION MANAGEMENT AND IAC

Systems Manager and AppConfig

Managing fleets as one: commands, desired state, runbooks, patching, secure access, inventory, life after OpsWorks, and feature flags.

DOMAIN 2 of 6 EXAM WEIGHT 17% MODULE 13 of 34

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The agent is the hand on every node. A document is the instruction. Targets decide which hands move, and when.

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Which Systems Manager tool?

THE NEED	TOOL
Run something on many nodes, once, now	Run Command
Keep nodes in a desired state, continuously	State Manager
A multi-step operational procedure, with approvals	Automation
Scan and install patches, and report compliance	Patch Manager
Shell access without SSH keys or open ports	Session Manager
Know what software is installed, everywhere	Inventory
Install or update software packages	Distributor

ON THE EXAM

Once: Run Command. Always: State Manager. A procedure: Automation. The verb in the scenario picks the tool.

What makes a node 'managed'

- The **SSM Agent** is installed and running. It is preinstalled on many AWS AMIs.
- The node has **permissions**: an instance profile with AmazonSSMManagedInstanceCore, or DHMC.
- The node has a **network path**: internet access, or interface endpoints for ssm, ssmmessages and ec2messages.
- On-premises and multicloud servers join through a **hybrid activation**.

> ON THE EXAM
>
> Node missing from Systems Manager: check agent, permissions, network path, in that order.

Default Host Management Configuration

- **DHMC** gives every EC2 instance in an account and Region Systems Manager permissions.
- No instance profile needed. Credentials reach the agent through the instance metadata service.
- Requires IMDSv2 and a recent agent version.
- Enable it across an organisation with Quick Setup.

> ON THE EXAM
>
> Instances launched by many teams without instance profiles must all be managed: turn on DHMC, not per-team profiles.

Run Command, or State Manager?

STATE MANAGER

Keep it that way

- An association: document, targets, schedule
- Re-applied on a schedule, and to new nodes
- Reports association compliance
- Corrects drift automatically

RUN COMMAND

Do it now

- One execution, on the targets of the moment
- Nodes launched later are not included
- Rate controls: concurrency and error threshold
- Output to S3 or CloudWatch Logs

THE TIEBREAKER

'Ensure', 'continuously', 'including new instances': State Manager. 'Now', 'once', 'urgently': Run Command.

Useful AWS-owned documents

DOCUMENT	WHAT IT DOES
AWS-RunPatchBaseline	Scans for or installs patches against a baseline
AWS-ApplyChefRecipes	Runs Chef cookbooks from S3 or Git, no Chef server
AWS-ApplyAnsiblePlaybooks	Runs Ansible playbooks on the node
AWS-GatherSoftwareInventory	Collects inventory data
AWS-ConfigureAWSPackage	Installs a Distributor package, such as the CloudWatch agent
AWS-UpdateSSMAgent	Keeps the agent itself current

ON THE EXAM

Before writing a custom document, look for an AWS-owned one. Exam answers usually use them.

Life after OpsWorks

RETIRED	ENDED	REPLACE WITH
OpsWorks Stacks	26 May 2024	State Manager with AWS-ApplyChefRecipes
OpsWorks for Chef Automate	5 May 2024	Chef cookbooks through State Manager
OpsWorks for Puppet Enterprise	31 March 2024	State Manager, or a self-run Puppet server

ON THE EXAM

Run existing Chef cookbooks periodically, with no Chef server to manage: a State Manager association with AWS-ApplyChefRecipes.

Automation runbooks

- Multi-step procedures: call APIs, run scripts, run commands on nodes, branch, wait, approve.
- Hundreds of AWS-owned runbooks, such as AWS-RestartEC2Instance and AWSSupport troubleshooting runbooks.
- Run under an **assume role**, with rate control, across accounts and Regions.
- Started by hand, by EventBridge rules, by Config remediation, or by maintenance windows.

> ON THE EXAM
>
> Automation is the usual 'target' that makes an event-driven fix happen. Domain 5 relies on it heavily.

A runbook with a human in the loop

```yaml
schemaVersion: '0.3'
assumeRole: '{{ AutomationAssumeRole }}'
parameters:
  InstanceId: { type: String }
  AutomationAssumeRole: { type: String }
mainSteps:
  - name: approve
    action: aws:approve
    inputs: { Approvers: ['arn:aws:iam::111122223333:role/OnCallLead'] }
  - name: restart
    action: aws:executeAwsApi
    inputs: { Service: ec2, Api: RebootInstances, InstanceIds: ['{{ InstanceId }}'] }
```

8 Pause until an approver approves or rejects.

11 Then call the EC2 API directly: no script, no agent needed.

ON THE EXAM

aws:executeAwsApi calls any API; aws:executeScript runs Python or PowerShell; aws:runCommand runs on nodes.

Patch Manager

- A **patch baseline** says which patches are approved: rules with auto-approval delays, plus approved and rejected lists.
- AWS-RunPatchBaseline runs **Scan** to report, or **Install** to patch, with a reboot option.
- Compliance reports which nodes are missing which patches.
- Each scan overwrites the previous compliance result. Keep history elsewhere if auditors need it.

ON THE EXAM

'Approve security patches seven days after release' is an auto-approval rule in a custom baseline.

Patch policies, or maintenance windows?

PATCH POLICY

Quick Setup, organisation-wide

- One policy across OUs and Regions
- Separate scan and install schedules
- Applies to new accounts and nodes
- The recommended approach today

MAINTENANCE WINDOW

The classic, per account

- A schedule with targets and tasks
- Patch groups chosen by tag
- Set up in each account and Region
- Also runs other tasks, not just patching

THE TIEBREAKER

Patch many accounts and Regions with least overhead: a Quick Setup patch policy targeting OUs.

Scheduling safely

- **Maintenance windows** run tasks, such as commands or runbooks, in a defined period.
- **Change Calendar** defines periods when changes are blocked, such as a trading freeze.
- Automation, State Manager and maintenance windows can check the calendar before acting.
- **Change Manager** closed to new customers on 7 November 2025.

> ON THE EXAM
>
> Stop automated changes during a freeze without editing every schedule: a Change Calendar.

Session Manager

- Shell or PowerShell access through the agent: **no inbound ports, no SSH keys, no bastions**.
- Access is controlled by IAM. Sessions can be logged to S3 or CloudWatch Logs, and encrypted with KMS.
- Port forwarding and SSH tunnelled through Session Manager are **not** content-logged.
- **Just-in-time node access** (April 2025) adds approval policies and time-bound access.

> ON THE EXAM
>
> Just-in-time access only protects nodes once users' direct StartSession permission has been removed.

Inventory, Distributor and Fleet Manager

- **Inventory** collects installed software, files, services and more from every node.
- **Resource Data Sync** sends inventory from many accounts and Regions to one S3 bucket, for Athena queries.
- **Distributor** packages software, yours or AWS's, and installs versions across the fleet.
- **Fleet Manager** is a console for browsing files, logs and performance on nodes.

> ON THE EXAM
>
> Which instances in every account run a vulnerable library version? Inventory plus Resource Data Sync plus Athena.

AWS AppConfig

- Deploys **feature flags** and configuration to running applications, **without redeploying code**.
- **Validators**, a JSON schema or a Lambda function, reject bad configuration before it deploys.
- **Deployment strategies**: linear or exponential growth, then a **bake time** watched by alarms.
- A CloudWatch alarm during deployment or bake triggers **automatic rollback**.

> ON THE EXAM
>
> CloudWatch Evidently was discontinued in October 2025. AppConfig now covers feature flags and experiments.

AppConfig, or Parameter Store?

APPCONFIG

Change behaviour safely

- Validated before deployment
- Gradual rollout with bake time
- Rolls back on alarms
- Feature flags and experiments

PARAMETER STORE

Store values

- A value changes immediately for every reader
- No rollout, no validation, no rollback
- Secrets as SecureString
- Great for configuration read at start-up

THE TIEBREAKER

Turn a feature on gradually and roll back on errors: AppConfig. Store a connection string: Parameter Store.

A company runs Linux and Windows instances in 30 accounts across two Regions, organised in AWS Organizations. Patching is currently done by hand.

The company wants patch compliance scanned daily, security patches installed weekly, new accounts covered automatically, and the least operational overhead.

Which solution meets these requirements?

A. A cron job on every instance that runs the operating system's package manager weekly

B. Maintenance windows running AWS-RunPatchBaseline, created in each account and Region

C. A Quick Setup patch policy targeting the organisation's OUs and both Regions, with daily scans and weekly installs

D. A Lambda function that loops through every account and runs patch commands

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A Cron on each instance

No compliance reporting, nothing for Windows, and every new instance must be configured.

B Maintenance windows everywhere

Works, but sixty set-ups to build and maintain, and new accounts are not covered.

C Quick Setup patch policy

One policy across OUs and Regions, separate scan and install schedules, new accounts included.

D Looping Lambda

Custom code rebuilding what patch policies already provide.

THE SIGNAL

Many accounts and Regions, new accounts automatic, least overhead: a patch policy.

> Engineers reach private EC2 instances through a bastion host using shared SSH keys. A security review requires removing the bastion, the keys, and all inbound ports.
>
> Engineers still need interactive shell access, and every command they type must be recorded for audit.

Which solution meets these requirements?

A. Keep the bastion, but require MFA and rotate the SSH keys monthly

B. An EC2 Instance Connect Endpoint with short-lived SSH keys

C. A client VPN into the VPC, then SSH to the instances

D. Session Manager with session logging to S3 or CloudWatch Logs, and port 22 closed

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A **Bastion with MFA**

Keeps the bastion and the keys the review said to remove.

B **Instance Connect Endpoint**

Removes the bastion, but is still SSH to port 22, and does not record commands.

C **Client VPN and SSH**

Keys and inbound SSH remain. Nothing is recorded.

D **Session Manager with logging**

No bastion, no keys, no inbound ports, and session content logged for audit.

THE SIGNAL

No keys, no ports, every command recorded: Session Manager with logging. Avoid its tunnels if content must be logged.

> In a new account, dozens of teams launch EC2 instances from their own launch templates. Most templates attach no instance profile, so the instances never appear in Systems Manager.
>
> The platform team wants every instance in the account and Region managed by Systems Manager, without changing each team's templates.

What should the platform team do?

A Ask every team to add an instance profile to its launch templates

B Install the SSM Agent through user data in every launch template

C Enable Default Host Management Configuration in the account and Region

D Create interface VPC endpoints for Systems Manager

◆ PAUSE AND DECIDE

What should the platform team do?

A Instance profiles everywhere

Works, but changes every team's template, which the requirement ruled out.

B Agent via user data

The agent is usually present already. Permissions are what is missing.

C Enable DHMC

Gives every instance in the account and Region the permissions it lacks, with no template changes.

D VPC endpoints

Fixes network paths, not missing permissions. The instances still cannot register.

THE SIGNAL

Agent present, no instance profile, many teams: default host management, once per account and Region.

> A retailer has built a new checkout flow, already deployed but switched off. It wants to enable it for a growing share of users over an hour, without redeploying code.
>
> If the checkout error rate alarm fires during the rollout, or in the half hour afterwards, the feature must switch off automatically.

Which solution meets these requirements?

A Change a Lambda environment variable to turn the feature on

B Store the flag in Parameter Store and have the application poll it

C A CodeDeploy canary deployment of a new version with the flag on

D An AppConfig feature flag with a linear deployment strategy, a 30-minute bake time, and the alarm attached

◆ PAUSE AND DECIDE

Which solution meets these requirements?

A Environment variable

Everyone at once, a function update, and no automatic rollback.

B Parameter Store flag

No gradual rollout and no alarm rollback. Everyone sees the change at once.

C CodeDeploy canary

Gradual and alarm-aware, but it redeploys code, which was ruled out.

D AppConfig feature flag

Gradual rollout, bake time, and alarm rollback, with no code deployment.

THE SIGNAL

Gradual, reversible, no redeploy: an AppConfig flag. The same safety ideas as Domain 1, applied to configuration.

Module 13, and Domain 2, in five lines

- Managed node: agent, permissions, network path. DHMC supplies permissions account-wide.
- Run Command once; State Manager continuously; Automation for procedures with approvals.
- OpsWorks is gone: Chef cookbooks run through State Manager. Patch policies patch organisations.
- Session Manager replaces bastions and keys; just-in-time access adds approvals.
- AppConfig rolls configuration out gradually and rolls back on alarms.

> ON THE EXAM
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> Domain 2 complete. Next, module 14 opens Domain 3 with high availability and multi-Region design.